

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE:ITA0604

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:100

Annexure A

Analytical Problem Solving

Code of Conduct:

I Jeevan Raaja S (192421005) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

11. Perceptron Model for Customer Purchase Prediction

Introduction

A perceptron is one of the simplest types of artificial neural networks used for binary classification. It determines whether an input belongs to one of two classes by calculating a weighted sum of the input features, adding a bias, and applying a step activation function. In this problem, the perceptron is used to predict whether a customer will purchase a product based on advertisement exposure and income level.

Given Data

- Weight for advertisement exposure, $w_1 = 0.5$
- Weight for income level, $w_2 = 0.3$
- Bias, $b = -0.4$
- Advertisement exposure, $x_1 = 2$
- Income level, $x_2 = 1$

Formula for Net Input

The net input to the perceptron is calculated using the formula:

$$\text{Net Input} = (w_1 \times x_1) + (w_2 \times x_2) + b$$

Step 1: Calculate the Net Input

Substitute the given values into the formula:

$$\begin{aligned} \text{Net Input} &= (0.5 \times 2) + (0.3 \times 1) + (-0.4) \\ \text{Net Input} &= (0.5 \times 2) + (0.3 \times 1) + (-0.4) = 1.0 + 0.3 - 0.4 = 1.0 + 0.3 - 0.4 = 0.9 \end{aligned}$$

Therefore,

$$\text{Net Input} = 0.9$$

Step 2: Apply the Step Activation Function

The step activation function is defined as:

- If $\text{Net Input} \geq 0$, $\text{Output} = 1$
- If $\text{Net Input} < 0$, $\text{Output} = 0$

Since,

$$0.9 \geq 0$$

the output of the perceptron is:

Output=1\boxed{\text{Output} = 1}Output=1

Explanation

The positive net input indicates that the combined effect of the customer's advertisement exposure and income level is strong enough to activate the perceptron. Hence, the model predicts that the customer is likely to purchase the product.

Conclusion

The calculated net input is 0.9, and after applying the step activation function, the output is 1. Therefore, the perceptron predicts that the customer will purchase the product.

12. Weight Update Using Delta Learning Rule

Introduction

The Delta Learning Rule is a supervised learning algorithm used to train a neuron by minimizing the error between the desired output and the actual output. During training, the weights are adjusted in the direction that reduces the prediction error.

Given Data

- Initial weight, $w = 0.6$
- Learning rate, $\eta = 0.15$
- Input, $x = 2.5$
- Actual output, $y = 0.7$
- Target output, $t = 1$

Step 1: Calculate the Error

The error is calculated as:

$$\text{Error} = \text{Target Output} - \text{Actual Output} \quad \text{Error} = \text{Target Output} - \text{Actual Output}$$

Substituting the values,

$$\text{Error} = 1 - 0.7 = 0.3 \quad \text{Error} = 1 - 0.7 = 0.3$$

Therefore,

$$\text{Error} = 0.3$$

Step 2: Calculate the Weight Update

The weight update is calculated using the Delta Rule:

$$\Delta w = \eta \times \text{Error} \times x \quad \Delta w = \eta \times \text{Error} \times x$$

Substituting the given values,

$$\Delta w = 0.15 \times 0.3 \times 2.5 \quad \Delta w = 0.15 \times 0.3 \times 2.5 = 0.1125 = 0.1125$$

Therefore,

$$\text{Weight Update} = 0.1125$$

Step 3: Calculate the Updated Weight

The updated weight is calculated as:

$$w_{\text{new}} = w + \Delta w \quad w_{\text{new}} = w + \Delta w$$

Substituting the values,

$$w_{\text{new}} = 0.6 + 0.1125 \quad w_{\text{new}} = 0.6 + 0.1125 = 0.7125 = 0.7125$$

Therefore,

$$\text{Updated Weight} = 0.7125$$

Explanation

The positive error indicates that the actual output is lower than the desired output. Therefore, the weight is increased by 0.1125, enabling the neuron to produce an output closer to the target value during the next iteration.

Conclusion

After one iteration of the Delta Learning Rule:

- Error = 0.3
- Weight Update = 0.1125
- Updated Weight = 0.7125

The neuron successfully adjusts its weight to reduce the prediction error, thereby improving its learning performance.

13. Performance Evaluation Using Confusion Matrix

Introduction

A confusion matrix is a performance evaluation tool used in machine learning classification problems. It compares the actual class labels with the predicted class labels and helps calculate important evaluation metrics such as Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-Score.

Given Data

- True Positives (TP) = 140
- True Negatives (TN) = 180
- False Positives (FP) = 25
- False Negatives (FN) = 35

Total number of observations:

$$\begin{aligned} \text{Total} &= \text{TP} + \text{TN} + \text{FP} + \text{FN} \\ &= 140 + 180 + 25 + 35 = 380 \end{aligned}$$

Step 1: Calculate Accuracy

Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Substituting the values,

$$\begin{aligned} \text{Accuracy} &= \frac{140 + 180}{380} \\ &= \frac{320}{380} = 0.8421 \\ &= 84.21\% \end{aligned}$$

Step 2: Calculate Precision

Formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Substituting the values,

$$\begin{aligned} \text{Precision} &= \frac{140}{140 + 25} \\ &= \frac{140}{165} = 0.8485 \\ &= 84.85\% \end{aligned}$$

Step 3: Calculate Recall (Sensitivity)

Formula:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Substituting the values,

$$\begin{aligned} \text{Recall} &= \frac{140}{140 + 35} \\ &= \frac{140}{175} = 0.80 \\ &= 80\% \end{aligned}$$

Step 4: Calculate Specificity

Formula:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

Substituting the values,

$$\text{Specificity} = \frac{180}{180 + 25} = \frac{180}{205} = 0.8780 = 87.80\%$$

Step 5: Calculate F1-Score

Formula:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Substituting the calculated values,

$$\text{F1-Score} = \frac{2 \times 0.8485 \times 0.80}{0.8485 + 0.80} = \frac{1.3576}{1.6485} = 0.8237 = 82.37\%$$

Summary of Results

Performance Metric	Calculated Value
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Accuracy	84.21%
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Precision	84.85%
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Recall (Sensitivity)	80.00%
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Specificity	87.80%
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F1-Score	82.37%
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Analysis

The classification model demonstrates good overall performance, with an accuracy of 84.21%, indicating that it correctly classifies most of the instances. The precision of 84.85% shows that the majority of positive predictions are accurate, reducing the number of false alarms. A recall of 80% indicates that the model is capable of identifying a large proportion of actual positive cases, while the specificity of 87.80% reflects its strong ability to correctly identify negative cases. The F1-score of 82.37% indicates a balanced trade-off between precision and recall.

Suitability for Fraud Detection

In fraud detection applications, false positives are costly because they may incorrectly classify genuine customer transactions as fraudulent, leading to inconvenience and additional verification costs. Since the model has high precision (84.85%) and high specificity (87.80%), it produces relatively fewer false positives. Therefore, the model is reasonably suitable for fraud detection applications. However, improving precision even further would help reduce false positives and make the system more efficient in real-world financial environments.

Conclusion

The confusion matrix evaluation shows that the classification model performs effectively across all major performance metrics. With an accuracy of 84.21%, precision of 84.85%, recall of 80%, specificity of 87.80%, and an F1-score of 82.37%, the model demonstrates reliable and balanced classification performance. Overall, it is suitable for many practical applications, including fraud detection, where minimizing false positives is an important requirement.